

A data driven Information theoretic feature extraction in EEG-based Motor Imagery BCI

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Abstract— Motor Imagery (MI) is the most popular Brain-Computer Interface (BCI) model which aimed at analyzing and classifying the electroencephalogram (EEG) measured without direct human's motor movements. The EEG recording is measured on the scalp noninvasively, which has nonstationarity and nonlinearity. To tackle the obstacle for analyzing the EEG obtained during MI tasks, we propose a novel feature extraction method by combining the Hilbert-Huang Transform (HHT) and the dispersion entropy (DisEn). Here, we develop the multivariate HHT using intrinsic mode functions (IMFs) obtained through multivariate empirical mode decomposition (MEMD) instead of HHT using existing EMD. By comparing the classification performance with other traditional methods, we validate the improved capacity of the proposed method, which shows its usefulness in MI BCI model.

Keywords—Brain-Computer Interface, Motor Imagery, Electroencephalogram, multivariate Hilbert-Huang Transform, Dispersion Entropy

I. INTRODUCTION

To measure brain activity, invasive and non-invasive measures have been used, where invasive measures that connect the electrodes directly to the brain with surgical incisions and noninvasive ones that connect the electrodes to the scalp. Electroencephalography (EEG) is a standard non-invasive measure which reflects the flow of electrical current that occurs between cranial nerves exchanging information with each other, usually measured on the scalp. Because the EEG recordings are measured on the scalp, it is susceptible to noise such as muscle movements compared to other brain monitoring methods. However, the EEG recording is widely used in lots of neural engineering applications such as Brain-Computer Interface (BCI) research owing to ease of measurement of brain activity without additional surgery.

BCI is an emerging field in neural engineering, which provides various interaction tools between brain activities and several external devices such as robot arms, a wheelchair, and so on. Motor Imagery BCI (MI-BCI) is one of BCI models that detects and/or classifies the EEG recording while a subject is imaging motor movements without direct motor movements. In addition, MI-BCI is preferred due to the ease of implementation. However, it is known that the MI-BCI using EEG has been suffering from low classification performance and illiteracy for 20% of subjects.

Here, we propose a novel feature extraction method for classifying the left and right hand MI EEG. From a viewpoint

of neurophysiology, the EEG features of the left and right hand MI are revealed in the electrodes C3, Cz, and C4 which correspond motor cortex. In addition, MI features are contralateral, which means the opposite motor cortex reflects MI patterns. The widely used MI patterns in EEG is Event-related desynchronization (ERD) and Event-related synchronization (ERS) shown in the mu band (8~13Hz) and beta band (13~30Hz) [1, 2]. ERD denotes the decrease of the energy appearing at the electrode at the start of MI and ERS describes the increase of energy at the electrode at the end of MI.

The Hilbert-Huang Transform (HHT) which is composed of the Empirical Mode Decomposition (EMD) and Hilbert transform (HT) is capable of analyzing non-stationary and nonlinear signals [3]. EMD decomposes a signal into sub-components which is called intrinsic mode functions (IMFs) in a data-driven way. Using the obtained IMFs, Hilbert transform is performed to obtain a frequency-time distribution of spectrum of the signal. Compared to classical nonlinear system analysis method, HHT is effective for analysis of nonlinear and nonstationary signal [4]. In this study, we make use of the multivariate HHT using IMF obtained through MEMD instead of EMD to analyze the signals of multivariate variables.

To quantify the EEG time-series, we utilize an information theory-based entropy which captures irregularity of the time-series. The most commonly used entropy methods are Sample Entropy (SE) and Permutation Entropy (PE). However, the SE slows down the computation rate in proportion to the amount of data, and it is difficult to perform the real-time analysis. The PE has a disadvantage in that it does not take into consideration the difference between neighboring data. Here, we use Dispersion Entropy (DisEn) which overcomes conventional entropy measures. DisEn quantifies the pattern distribution of data as a way to analyze the dynamic change of data according to time-series [5].

In this paper, we propose a novel feature extraction method which consists of HHT and DisEn to analyze MI-EEG, thus dealing with nonstationary and nonlinear characteristics of EEG recordings. First, we obtain an energy distribution of the signal in the frequency-time domain through HHT with the IMFs. Then, DisEn is used to quantify the irregularity of the energy of EEG that changes over both time and frequency. The frequencies used are the Mu (8~13 Hz) and Beta (13~30 Hz) bands, which reflect ERD and ERS of MI-EEG. The classification performance of the proposed method is verified by using SVM with RBF kernel. The public BCI dataset, called BCI Competition II, is used in this study.

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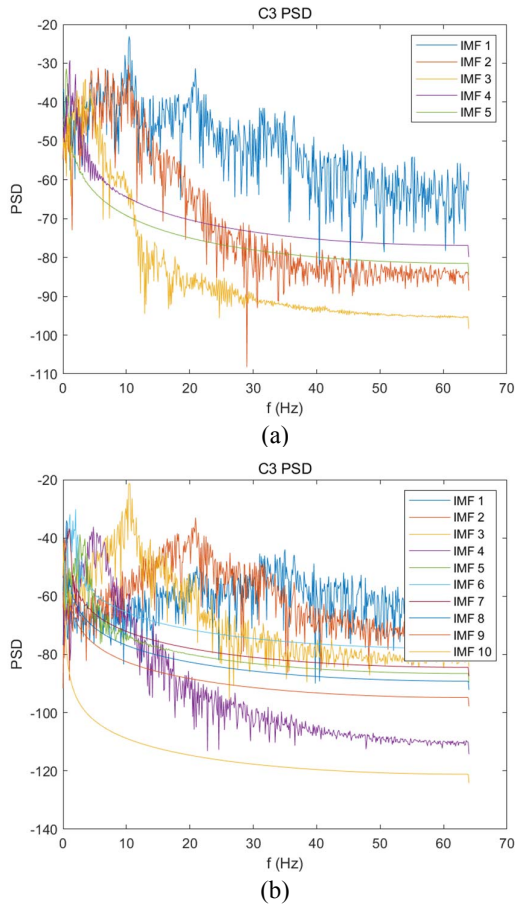


Fig. 2. It is the PSD value of IMF for electrode C3 obtained from EMD or MEMD. (a) PSD of IMF obtained through EMD. (b) PSD of IMF obtained through MEMD

The PSD of IMFs is calculated as shown in Fig. 2 in order to select the IMF which reflects characteristics of MI. Fig. 2 (a) depicts the PSD of the IMFs obtained by EMD and Fig. 2(b) shows that of the IMFs obtained by MEMD. As can be seen in Fig. 2, the IMFs after the 5th IMF have too low value of PSD in the range of 8~30 Hz. Therefore, we have combined IMFs 1~5 to create 15 new data sets. The ERD/ERS feature of the MI appears in the Mu bands (8~13 Hz), so the highest classification accuracy is expected in IMF1 and IMF2 for EMD and IMF3 for MEMD.

Table 1 shows the maximum accuracy obtained according to the IMF used for the HHT. We tested for four cases in terms of methods to derive IMFs and the number of channels. In addition, an embedding dimension m is set to integer values between 1 and 3, class c is set to values between 2 and 14. The time delay d is chosen as 1.

In recent work, Hilbert-Huang Entropy (HHE), which consists of EMD and SE was used for MI analysis [7]. It accomplished the accuracy of 85% when the first IMF and the second IMF were used. As shown in Table I, the combination of EMD and DisEn proposed in this paper shows 85% accuracy using the first and second IMFs, which are similar to the HHE method. Notably, there is no significant difference in cases of 2 and 3 channels which is used. It reveals that Cz electrode does not play a role in providing meaningful information on MI.

TABLE I. CLASSIFICATION ACCURACY USING THE PROPOSED METHODS

IMF	MEMD C3, C4	MEMD C3, Cz, C4	EMD C3,C4	EMD C3, Cz, C4
1	56.43	55.00	73.57	73.57
2	62.86	64.29	82.14	80.71
3	87.86	87.14	71.43	75.00
4	60.71	58.57	63.57	61.43
5	58.57	57.86	50.71	55.00
1~2	60.00	60.00	85.00	84.29
1~3	85.00	82.86	80.00	82.86
1~4	85.00	84.27	80.71	80.71
1~5	84.29	85.71	82.14	78.57
2~3	85.71	83.57	79.29	80.00
2~4	86.43	85.71	81.43	80.71
2~5	81.43	78.57	80.71	80.00
3~4	80.00	80.00	71.43	71.43
3~5	87.86	89.29	70.71	72.14
4~5	59.29	55.00	64.29	65.00

When using the first and second IMFs from MEMD and following DisEn, it yields to lower accuracy compared to other cases. However, other combinations of IMFs from MEMD, such as 1~3, 1~4 or 1~5 IMFs, lead to comparable accuracies to the traditional method.

Moreover, using a specific IMF combination, a third IMF or 3~5 IMFs provides the highest classification accuracy and superior to conventional method.

IV. CONCLUSION

To conclude, we have presented a novel way to discriminate MI-EEG by combining data-driven decomposition methods and following DisEn computation. The HHT using the IMFs obtained from EMD and MEMD maps the EEG recordings into the energy distribution in the frequency-time domain, and the time-series information of the specific frequency is captured via DisEn. To verify the proposed method, we classified the left and the right hand MI by using the EEG measured during the movement imagination. The proposed method combining multivariate HHT and DisEn shows outperforms conventional methods. These results reveal the capability of the proposed method for MI-EEG field and availability for being applied to other EEG signal processing applications.

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